Finite Packet Certificates for Near-Kerr Evolution: Beam Concentration, Derivative Remainders, and Physical Error Budgets

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M533

October 9, 2026

Abstract

We study the finite-signal construction in the assigned near-Kerr manuscript, without independently certifying its global evolution or generic future-inextendibility theorem. A quantitative concentration bound for projected beam centers replaces the source's absolute three-dimensional center-integration loss by a fractional concentration loss, including an explicitly sharp fold example. We prove differentiated Gaussian remainder bounds, trace the improvement through the exterior packet and data-seminorm budgets, and retain every additional geometric premise. An exact positive triangular propagator sharpens the finite mixed-energy certificate without repeating a common exponential rate at each derivative level. Finite complex-Hessian and Hamiltonian-root certificates distinguish algebraic positivity, symmetry, branch control, and actual propagation. These results require full derivative and unmeasured remainder bounds; polynomial jets alone provide none of the global vacuum, blueshift, or Cauchy-attachment inputs. Contribution assessment: C3/5, the concentration-sensitive packet and finite-window error certificates give a notable conditional local refinement while leaving the global near-Kerr and inextendibility claims imported.

Contents

1 The Source Problem and the Local Entry Point **2**
1.1 Architecture and the order of quantifiers 2
1.2 Normalized physical fields 3

2 Projected Center Concentration **3**
2.1 What must be differentiated 5
2.2 Geometric sufficient conditions and a sharp fold 5

3 A Conditional Improvement of the Packet Budget **6**

4 Complex Phase Safety with Approximate Blocks **7**

5 The Exact Finite Triangular Error Budget **8**
5.1 The physical bulk rate 9

6 A Finite Hamiltonian Branch Certificate **10**

7 Comparison, Limitations, and Further Work **11**
7.1 What is and is not improved 11
7.2 Primary public context checked on October 9, 2026 11
7.3 Feasible extensions 11

8 Verification and Evidence Boundaries **12**

9 Conclusion **12**

1 The Source Problem and the Local Entry Point

The source [1] fixes a rotating subextremal Kerr bridge, $M > 0$, $0 < \mathfrak{a} < M$, with $\Sigma \simeq \mathbb{R} \times \mathbb{S}^2$, background (h_*, K_*), connection ∇_*, and radial weight $\rho \geq 1$. Its seminorms are

$$p_m(u, v) = \sum_{j=0}^{m} \left(\sup_{\Sigma} \rho^{1+j} |\nabla_*^j u|_* + \sup_{\Sigma} \rho^{2+j} |\nabla_*^j v|_* \right). \tag{1.1}$$

The smooth vacuum constraint space consists of complete positive metrics and second forms satisfying

$$R(h) + (\mathrm{tr}_h K)^2 - |K|_h^2 = 0, \qquad \mathrm{div}_h K - \mathrm{d}(\mathrm{tr}_h K) = 0,$$

with all seminorms finite. The topology uses *all* these seminorms. The neighborhood asserted in the source is nevertheless a single p_{10}-ball. Higher seminorms need not be small, and neither end has prescribed leading asymptotic coefficients.

A future extension is a proper open time-oriented isometric embedding of the full maximal globally hyperbolic development into a connected Lorentzian manifold with a nondegenerate C^2 metric, reached at its boundary by a future timelike curve. The extension need not be vacuum outside the image. The source claims that extendible data are meagre in one such neighborhood, for each fixed mass and spin. There is no uniform claim as the spin approaches zero or extremality.

1.1 Architecture and the order of quantifiers

The original introduction, category argument, entire packet section, and selected adjacent proof passages were freshly read; the final read ledger gives the literal intervals and all unread complements. The architecture is:

1. The category section builds closed tests $A(O, N)$: every seed in an open set $O \subset T\Sigma$ has proper duration at most N and parallel-frame curvature norm at most N. A strict failure has a finite compact witness. A regular-exit covering and Baire argument reduce the claim to destroying these tests.

2. Geometry and stationary sections supply the Kerr ray geometry, directional ellipticity, gauges, and linear spectral inputs. Compact/far sections and their two long appendices assert the coupled evolution estimates for the arbitrary symbol tails.

3. The interior section supplies a full double-null wedge, low bounds, pure longitudinal jets, and higher estimates $e^{o(T)}$. The comparison section connects an arbitrary regular exit to a branch seed in the original open family.

4. The packet section constructs three different objects: a deep approximate packet h^p, an exterior beam superposition h^B, and exact constraint-corrected data d_T. Ordinary wave-map comparison with the actual background as target transfers the jets.

5. A protected-ball constraint inverse permits a noncompact charge-changing tail. A separate full-bridge Cauchy attachment places the finite observation in the new full development.

Most of the global stages in this list are imported, including their uniformity on one base neighborhood. Reading a displayed proof is not independent certification of all its prerequisites.

The finite derivative order m, comparison buffers, and positive nonlinear loss exponents are fixed *before* the frequency exponent. Formal accuracy is fixed afterwards, and late time T last. Here $e^{o(T)}$ means $C_\sigma e^{\sigma T}$ for each fixed $\sigma > 0$, with constants depending on the datum and fixed orders. It supplies no computable threshold and no uniformity as the derivative order grows.

1.2 Normalized physical fields

In the fixed background coordinates of the packet section,

$$u = y_0, \quad v = y_1, \quad s = (u+v)/2, \quad x = (v-u)/2,$$

$$g = -2a\ \mathrm{d}u\ \mathrm{d}v + \gamma_{AB}(\,\mathrm{d}\theta^A - b^A\,\mathrm{d}u)(\,\mathrm{d}\theta^B - b^B\,\mathrm{d}u), \qquad a = q_0 e^{-\kappa(u+v)} A.$$

Put $L_0 = \partial_u + b^A\partial_A$, $L_1 = \partial_v$, $W_t = L_0 + L_1$, $W_x = L_1 - L_0$, and normalized slots $\hat{e}_i = a^{-1/2}L_i$, $\hat{e}_A = \partial_A$. A two-tensor component with n_i radial i-slots has $c_n = \kappa(n_0 - n_1)/2$, hence $|c_n| \le \kappa$. The source's reference energy density is

$$\frac{1}{2}\left(|W_t f|^2 + |(W_x - 2c_n)f|^2 + 2a|d_\theta f|^2_\gamma + \varepsilon_0^2|f|^2\right)\,\mathrm{d}x\,\mathrm{d}\,\mathrm{vol}_\gamma\,. \tag{1.2}$$

Ordinary coordinate derivatives through a fixed order are summed in this norm. Neither $\mathrm{d}\,\mathrm{vol}_\gamma$ nor the lapse factors can be replaced by an unweighted coefficient norm.

At the late point $v = T$, the source establishes $Y_T = \dot{v} \asymp e^{\kappa T}$ on the chosen bounded-curvature branch. Its packet has

$$\lambda = e^{\zeta T}, \qquad A_T = \frac{4J_{\mathrm{amp}}}{Y_T^2\lambda^2\sigma_T} = e^{(-2\kappa-2\zeta)T+o(T)}, \tag{1.3}$$

where the screen polarization σ_T has two-sided subexponential size. Its leading tidal term is $Y_T^2 A_T \lambda^2 \sigma_T/2 = 2J_{\mathrm{amp}}$. Curvature of the background, packet remainder, exact-solution error, and frame normalization all still require control. The local improvements below do not change this normalization.

2 Projected Center Concentration

On a fixed finite lab slice let Y label beam centers in a subset of $\mathbb{R}^3$, let $X_t(Y)$ be their real positions, and let μ be their nonnegative center measure. In the source it is Lebesgue $\mathrm{d}Y$ in the chosen entry chart, not a probability law and not spatial volume. If amplitudes carry an additional measure density, its absolute value must be included in μ or in the amplitude bounds.

Assume, uniformly over the observation region and times being used,

$$\mu\{Y : |z - X_t(Y)| \le r\} \le M_T r^q \quad (r > 0), \qquad 0 \le q \le 3. \tag{2.1}$$

A small-radius bound extends to all radii by increasing M_T when the total center mass is finite. The case $q = 0$ is simply a total mass bound. A positive exponent is extra geometric information.

Lemma 2.1 (Gaussian moment budget)**.** *Let* (2.1) *hold and let* $\beta > 0$*. Writing* $r_Y = |z - X_t(Y)|$*, for* $d \geq 0$ *one has*

$$\int r_Y^d e^{-\beta r_Y^2} \, \mathrm{d}\mu(Y) \leq M_T C_{q,d} \beta^{-(q+d)/2}, \tag{2.2}$$

where

$$C_{q,0} = \Gamma(1 + q/2), \qquad C_{q,d} = \Gamma(1 + (q+d)/2) + \frac{d}{2}\Gamma((q+d)/2) \quad (d > 0).$$

For $d = 0$ *the constant is sharp among measures obeying the concentration hypothesis.*

Proof. For $d = 0$, write $e^{-\beta r^2} = \int_r^\infty 2\beta t e^{-\beta t^2} \, \mathrm{d}t$ and use Tonelli and (2.1). The result is $M_T\Gamma(1 + q/2)\beta^{-q/2}$. For $d > 0$, set $w(r) = r^d e^{-\beta r^2}$. Since $w(\infty) = 0$, $w(r) \leq \int_r^\infty |w'(t)| \, \mathrm{d}t$. Tonelli and

$$|w'(t)| \leq dt^{d-1}e^{-\beta t^2} + 2\beta t^{d+1} e^{-\beta t^2}$$

give the stated two gamma integrals. For sharpness when $q > 0$, use a radial measure with cumulative mass exactly $M_T r^q$, at the observation center. For $q = 0$, an atom of mass M_T at that center gives equality. These are sharpness fixtures for the measure inequality, not claims about a Kerr ray family. □

Theorem 2.2 (Differentiated absolute superposition)**.** *Suppose differentiation under the center integral is justified through order* j*, including required time derivatives. For each* $|\alpha| \leq j$*, assume the actual differentiated integrand obeys*

$$|D_{t,z}^\alpha b_\lambda(t, z, Y)| \leq H_{T,j} \lambda^{|\alpha|} e^{-c_T \lambda r_Y^2}, \qquad c_T > 0, \quad \lambda \geq 1.$$

Then for $\mathcal{B}_\lambda = (\lambda/2\pi)^{3/2} \int b_\lambda \, \mathrm{d}\mu$*,*

$$\left\| D_{t,z}^\alpha \mathcal{B}_\lambda \right\|_\infty \leq \frac{M_T H_{T,j} \Gamma(1 + q/2)}{(2\pi)^{3/2} c_T^{q/2}} \, \lambda^{|\alpha| + (3-q)/2}. \tag{2.3}$$

If a differentiated remainder instead obeys a finite bound

$$|D_{t,z}^\alpha b_\lambda^{\mathrm{rem}}| \leq \sum_\nu H_{\alpha,\nu} \lambda^{\ell_{\alpha,\nu}} r_Y^{d_{\alpha,\nu}} e^{-c_T \lambda r_Y^2}, \qquad d_{\alpha,\nu} \geq 0,$$

its integral is at most

$$\frac{M_T}{(2\pi)^{3/2}} \sum_\nu H_{\alpha,\nu} C_{q,d_{\alpha,\nu}} \, c_T^{-(q+d_{\alpha,\nu})/2} \lambda^{\ell_{\alpha,\nu} + (3-q-d_{\alpha,\nu})/2}. \tag{2.4}$$

Proof. Apply Lemma 2.1 with $\beta = c_T\lambda$ to each differentiated term, and retain the specified normalization. All norms may be taken componentwise in the declared normalized frame; conversion constants belong in H, not in an omitted convention. □

The finite bound (2.4) is the certificate: Taylor degree, derivative losses, Gaussian concentration, and omitted terms are visible separately. It does not assert that a finite jet is an exact solution of the tensor wave or gauge equations.

2.1 What must be differentiated

For a smooth amplitude a expanded about a fixed center X, put $h = z - X$. The multivariable integral remainder is

$$R_D a(z) = \frac{1}{D!} \int_0^1 (1-t)^D D^{D+1} a(X + th)[h^{D+1}] \, \mathrm{d}t.$$

If $|\beta| \le D + 1$, differentiation of the Taylor polynomial gives

$$D^\beta R_D a = D^\beta a - T_{D-|\beta|}(D^\beta a), \qquad |D^\beta R_D a| \le \frac{\sup \left\| D^{D+1} a \right\|}{(D + 1 - |\beta|)!} |h|^{D+1-|\beta|},$$

with the natural degree -1 convention at the last order. The derivative norm here bounds the relevant multilinear contractions. When the center moves in time, its differentiated motion contributes additional chain-rule terms and must be bounded.

For a phase ϕ, the exact product rule is

$$D^\alpha (e^{i\lambda\phi} R_D a) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^{\alpha-\beta}(e^{i\lambda\phi}) D^\beta R_D a.$$

Each phase derivative is a finite sum of products of actual phase jets and powers of λ. Those terms, and differentiated coefficients of the wave operator, must be inserted in (2.4). There is no rule deleting a coefficient derivative because its total formal label exceeds the amplitude degree. The original local-moment proof [2] gives Taylor's integral remainder; the original jet proofs [3] explicitly retain Leibniz rows and their differentiated remainders. These are classical analytic ingredients with a direct role here, not a hyperbolic propagator theorem.

A simple failure test is $R_\epsilon(x) = \epsilon \sin(x/\epsilon^2)$: its value tends uniformly to zero while its first derivative is unbounded. Pointwise matching therefore does not supply the $H^{k+1} \times H^k$ entry matching required by the source.

If $\operatorname{Im} \phi \ge c_T |z - X|^2$ on the actual tube and the needed jets are bounded, these product rules give the differentiated Gaussian premises of Theorem 2.2, after reducing c_T if needed to absorb polynomial factors. The source cuts off at radius $e^{-b_p T}$ with $0 < b_p < \zeta/2$. When $c_T^{-1} = e^{o(T)}$, the Gaussian exponent at that cutoff is

$$c_T \lambda e^{-2b_p T} = \exp((\zeta - 2b_p)T - o(T)).$$

It dominates every fixed exponential differentiation cost. Thus cutoff derivatives can be separately made smaller than every prescribed inverse power of λ. This argument needs the actual imaginary-phase lower bound throughout the cutoff annulus, not just positivity of its Hessian at the center.

2.2 Geometric sufficient conditions and a sharp fold

Proposition 2.3 (Regular projections and a fold)**.** *A finite collection of injective center charts with* $|\det D_Y X_t| \ge j_T > 0$ *and center density at most* W_T *gives* $q = 3$ *and* $M_T \le L_T W_T j_T^{-1} |B_1|$*, where* L_T *is the number of charts. Uniform bounds are required on the whole slice family. For*

$$X(Y) = (Y_1, Y_2, Y_3^2)$$

and Lebesgue center measure on a cube, one instead has $q = 5/2$ *with* $M = 8\sqrt{2}$*. In general* $q = 3$ *cannot hold near this fold.*

Proof. The regular statement is change of variables on each injective chart, followed by summation; overlaps can only increase that upper bound. For the fold, the first two preimage coordinates lie in intervals of length $2r$. The length of $\{y : |y^2 - z_3| \le r\}$ is at most $2\sqrt{2r}$. Their product is $8\sqrt{2}r^{5/2}$. Intersecting with the cube only decreases the measure. At $z = 0$, a rectangular subset with $|Y_1|, |Y_2| \le r/2$ and $|Y_3| \le \sqrt{r/2}$ lies in the radius-r ball preimage. Its volume is a fixed positive multiple of $r^{5/2}$, which rules out a uniform Mr^3 bound as $r \downarrow 0$. □

More generally (Y_1, Y_2, Y_3^ℓ), $\ell \ge 1$, gives $q = 2 + 1/\ell$, with a constant depending on ℓ. Indeed a length-$2r$ interval in the last image coordinate has preimage length at most $2(2r)^{1/\ell}$; the same box argument applies. This is a local finite-type model, not an assertion that every Kerr caustic has this form.

Example 2.4 (The power loss is real). On the full model center space $\mathbb{R}^3$,

$$\left(\frac{\lambda}{2\pi}\right)^{3/2} \int_{\mathbb{R}^3} \exp\left(-\frac{\lambda}{2}(Y_1^2 + Y_2^2 + Y_3^{2\ell})\right) \mathrm{d}Y \tag{2.5}$$

$$= \frac{2^{1/(2\ell)}\Gamma(1/(2\ell))}{\ell\sqrt{2\pi}}\, \lambda^{1/2 - 1/(2\ell)}. \tag{2.6}$$

This follows by two Gaussian integrals and the substitution $u = \lambda Y_3^{2\ell}/2$. On a fixed cube about zero the same leading asymptotic holds by rescaling and dominated convergence. For $\ell = 2$, the constant is $\Gamma(1/4)/(2^{5/4}\sqrt{\pi})$ and the power is $1/4$. Thus an absolute pointwise bound with no frequency loss is false at a fold, even though a dimension-free energy-norm estimate may still hold by oscillatory cancellation.

3 A Conditional Improvement of the Packet Budget

The source's exterior estimate is

$$\left\|h^B\right\|_{C^{N_b}} \le A_T \lambda^{N_b + 3/2} e^{o(T)}, \qquad \mathrm{Res} \le e^{o(T)}(A_T \lambda^{-M_1} + A_T^2 \lambda^{P_0}), \quad P_0 = 2N_b + 7.$$

The $3/2$ is attributed explicitly to absolute center integration in packet lines 947–960. We improve that particular bound under additional measured or proved geometric information.

Theorem 3.1 (Concentration-sensitive exponent budget)**.** *Retain the source's normalized wave/gauge hierarchies, physical comparison norms, constraint correction, and fixed derivative buffers. Assume on every used exterior computing slice that* (2.1) *holds with fixed $q \in [0, 3]$, and that $M_T, c_T^{-1}, H_{T,j} = e^{o(T)}$ at all required fixed orders. Assume the differentiated beam remainders meet* (2.4) *to arbitrarily prescribed finite inverse accuracy. For the nonlinear estimate, assume in the actual normalized arrays a conservative bilinear domination by derivatives through $N_b + 2$, with subexponential coefficients, while higher products are absorbed under the stated smallness bootstrap. Then one may replace the exterior bounds by*

$$\left\|h^B\right\|_{C^{N_b}} \le A_T \lambda^{N_b + (3-q)/2} e^{o(T)}, \tag{3.1}$$

$$\mathrm{Res} \le e^{o(T)} \left(A_T \lambda^{-M_1} + A_T^2 \lambda^{P_0 - q}\right). \tag{3.2}$$

The corresponding direct beam term in the prescribed data seminorm is $A_T \lambda^{m + (5-q)/2} e^{o(T)}$, with the same corrected-data and global-existence obligations as in the source.

Proof. Apply (2.3) through $N_b + 2$; all stated geometric and differentiated-amplitude factors are subexponential at these fixed orders. Squaring the conservative $N_b + 2$ bound gives the exponent $2N_b + 4 + (3 - q) = P_0 - q$. The normalized nonlinear domination is an explicit premise: this calculation is not a substitute for verifying the inverse-metric, gauge, and lapse placements in that bound. For each finite list of differentiated linear defects, increase the finite Taylor degree and hierarchy length until every exponent in (2.4) is at most $-M_1$. Cutoff errors were treated separately above. This proves (3.2). The second form needs one more beam derivative than the metric. Thus the direct order-m term costs $A_T \lambda^{m+1+(3-q)/2}$. The physical annular weights, rescaling, and volume conversions in the source cost only fixed powers of T, already included in $e^{o(T)}$. The exterior correction acts on the improved residual but is not reproved here. □

A verified $q > 0$ therefore weakens two explicit frequency restrictions. Keeping the source's entry target $e^{-7\kappa T/2}$ and its deep exponent P_*, sufficient displayed margins become

$$\begin{aligned} \delta_3 + (P_0 - q - 4)\zeta &< \kappa/2, & \delta_2 + 2\zeta &< \kappa/2, \\ \delta_2 + (P_* - 2)\zeta &< \kappa, & (m + (1-q)/2)\zeta &< \kappa, \\ (M_1 + 2)\zeta &> 3\kappa/2 + \delta_3, & M_1\zeta &> \kappa + \delta_2. \end{aligned} \tag{3.3}$$

For example the two exterior exponents are $-2\kappa - (M_1 + 2)\zeta + \delta_3$ and $-4\kappa + (P_0 - q - 4)\zeta + \delta_3$. Dividing the deep error by A_T leaves the same three exponents as in the source:

$$-\kappa/2 + \delta_2 + 2\zeta, \quad \kappa + \delta_2 - M_1\zeta, \quad -\kappa + \delta_2 + (P_* - 2)\zeta.$$

The improved direct data exponent is $-2\kappa + (m + (1 - q)/2)\zeta$. These substitutions prove (3.3); other base-norm absorption restrictions still have to be checked.

Example 3.2 (A strictly enlarged displayed parameter window). Take $\kappa = 1, \delta_2 = \delta_3 = 1/20, N_b = 20, P_* = 40, m = 10$, $q = 5/2, \zeta = 11/1000, M_1 = 150$. All six inequalities in (3.3) hold. The first left side is $991/2000 < 1/2$, whereas the source's $q = 0$ bound gives $523/1000 > 1/2$. This is an exact algebraic fixture of the displayed budget, not a verification of the geometric concentration assumption or all quasilinear bootstrap restrictions for an actual Kerr datum.

No change in the claimed global regularity or genericity conclusion follows from this local gain. If no positive concentration exponent is certified, $q = 0$ reproduces the original integration power. A finite collection of sampled rays does not prove (2.1) at arbitrarily small radii.

4 Complex Phase Safety with Approximate Blocks

The source propagates complex position and momentum blocks P, Q and uses an exact symplectic identity to keep the imaginary Hessian positive. That identity is inherited, not a new result here. A finite approximation requires two distinct defect checks. For matrices in this section, $\operatorname{Im} A = (A - A^*)/(2i)$ is the Hermitian imaginary part, not entrywise imaginary part. For scalar phases, $\operatorname{Im}\phi$ has its usual scalar meaning.

Proposition 4.1 (Hermitian and bilinear defects). *Let $P, Q \in \mathbb{C}^{d\times d}$, with Euclidean Hermitian norms, and put*

$$H = \frac{P^*Q - Q^*P}{2i}, \qquad B = P^TQ - Q^TP.$$

If $H \succeq hI$, $h > 0$, then P is invertible and

$$\sigma_{\min}(P) \geq \frac{h}{\|Q\|}, \qquad \operatorname{Im}(QP^{-1}) = P^{-*}HP^{-1} \succeq \frac{h}{\|P\|^2} I. \tag{4.1}$$

Writing $M = QP^{-1}$, one also has

$$M - M^T = P^{-T} B P^{-1}, \qquad \left\| M - M^T \right\| \le \frac{\|Q\|^2}{h^2} \|B\| .$$

If $M_s = (M + M^T)/2$, then

$$\operatorname{Im} M_s \succeq \mu I, \qquad \mu = \frac{h}{\|P\|^2} - \frac{\|Q\|^2}{2h^2} \|B\| . \tag{4.2}$$

Proof. For every v,

$$h|v|^2 \le v^* H v = \operatorname{Im}((Pv)^* Q v) \le |Pv| \, \|Q\| \, |v|.$$

This proves invertibility and the first bound. Multiplying H on the left by P^{-*} and the right by P^{-1} proves the second identity and inequality. The bilinear identity follows using transposes, not conjugate transposes. Finally $\|M_s - M\| \le \left\| M - M^T \right\| /2$, and the norm of the Hermitian imaginary part of a matrix is at most its norm. □

For example $P = I, Q = iI + J$ with a small nonzero real skew-symmetric matrix J has $H = I - iJ \succ 0$ but $B = 2J \neq 0$. Thus positivity alone does not prove the symmetry required of a phase Hessian. A lower bound $H \succeq H_0 - \delta I$ may be used with $h = \lambda_{\min}(H_0) - \delta > 0$. Errors in the computed blocks must first be propagated into this inequality.

Suppose an actual symmetric phase has zero imaginary value and gradient at its real center, imaginary Hessian at least μI, and $\|D^3 \operatorname{Im} \phi\| \le L_3$ on its tube. Taylor's theorem gives

$$\operatorname{Im} \phi(z) \ge \frac{\mu}{4} |z - X|^2 \quad \text{if} \quad |z - X| \le \frac{3\mu}{2L_3}, \tag{4.3}$$

with no radius restriction from this bound if $L_3 = 0$. Indeed the quadratic contribution is at least $\mu|z - X|^2/2$ and the remainder at most $L_3|z - X|^3/6$. Symmetrizing a computed matrix changes the phase model; its wave/gauge residual and higher jets must then be recomputed or enclosed. These finite algebraic checks supply neither the actual Hamiltonian flow nor the projected-center concentration bound.

5 The Exact Finite Triangular Error Budget

The packet proof uses square-root mixed energies E_j obeying

$$E_j' \le r E_j + b \sum_{\ell < j} E_\ell + S_j, \qquad 0 \le j \le k, \tag{5.1}$$

with nonnegative integrable coefficients and sources. Its appendix bounds the propagator by $\mathcal{Q}_k(B) = \sum_{\nu=0}^{k} ((k+1)B)^\nu / \nu!$. The following exact certificate keeps the same physical differential inequalities and source terms.

Theorem 5.1 (Sharp finite-order propagator)**.** *Let $E_j \ge 0$ be absolutely continuous and satisfy (5.1). Put $R(t) = \int_0^t r$, $B(t) = \int_0^t b$. For $x \ge 0$, define*

$$U_{j\ell}(x) = \begin{cases} 0, & \ell > j, \\ 1, & \ell = j, \\ \displaystyle\sum_{\nu=1}^{j-\ell} \binom{j-\ell-1}{\nu-1} \frac{x^\nu}{\nu!}, & \ell < j. \end{cases}$$

Then

$$E_j(t) \le e^{R(t)} \sum_{\ell \le j} U_{j\ell}(B(t)) E_\ell(0) + \int_0^t e^{R(t)-R(s)} \sum_{\ell \le j} U_{j\ell}(B(t)-B(s)) S_\ell(s)\,\mathrm{d}s. \tag{5.2}$$

The exact ℓ^∞ propagator norm is

$$\mathcal{L}_k(x) = \sum_{\nu=0}^{k} \binom{k}{\nu} \frac{x^\nu}{\nu!} \le \mathcal{Q}_k(x). \tag{5.3}$$

Neither the row coefficients nor this norm can be decreased for all systems satisfying these premises.

Proof. Let $A_{j\ell} = 1$ if $\ell < j$, zero otherwise. A length-ν increasing chain from ℓ to j chooses $\nu - 1$ intermediate indices out of $j - \ell - 1$. Thus $(A^\nu)_{j\ell} = \binom{j-\ell-1}{\nu-1}$, and $A^{k+1} = 0$. Removing e^R leaves the positive propagator $e^{(B(t)-B(s))A} = U(B(t) - B(s))$. Integration of the inequalities successively in j proves (5.2). The row sum of A^ν is $\binom{j}{\nu}$, so the largest exponential row sum is $\mathcal{L}_k$. Equality systems with nonnegative initial data and sources attain the coefficients; initial data all one and zero source attain the norm. Finally $\binom{k}{\nu} \le (k+1)^\nu$. □

This is a classical Laguerre expression, not a new special function. Its use here retains the actual derivative hierarchy and injection time of each residual. For $k = 3, B = 2$, $\mathcal{L}_3(2) = 43/3$, whereas $\mathcal{Q}_3(2) = 379/3$. This improves a certified bound, not a measured computation time. At fixed k, the source's $e^{o(T)}$ conclusion is unchanged. The auxiliary estimate

$$\mathcal{L}_k(B) \le \sum_{\nu \ge 0} \frac{(kB)^\nu}{(\nu!)^2} \le e^{2\sqrt{kB}}$$

follows from $\binom{k}{\nu} \le k^\nu/\nu!$ and $\binom{2\nu}{\nu} \le 4^\nu$. It does not supply the missing order-uniform beam or background estimates for growing k.

5.1 The physical bulk rate

For one real normalized component, put $X = W_t f,\ Y = (W_x - 2c)f,\ Z = \varepsilon_0 f$. The source's limiting nonsource production is

$$-4cXY + \varepsilon_0 XZ - 2\kappa a |d_\theta f|^2_\gamma.$$

The symmetric matrix of the first two terms has largest eigenvalue $\sqrt{4c^2 + \varepsilon_0^2/4}$: it couples X to $-2cY + \varepsilon_0 Z/2$. Since the reference density is half the squared norm, the square-root energy rate is at most

$$r_c = \sqrt{4c^2 + \varepsilon_0^2/4}. \tag{5.4}$$

This is sharp for the displayed quadratic form, not a saturating Einstein solution. A bulk excess rate at most $\delta > 0$ above 2κ follows from $\varepsilon_0 \le 2\sqrt{\delta(4\kappa + \delta)}$. Coefficient defects, nonbulk time, commutators and boundary fluxes must still be included in r, b, S. The source's exact-principal stress energy and strict outflow conditions are not replaced by this matrix calculation.

6 A Finite Hamiltonian Branch Certificate

The source traces the observed state backwards with v as time. On a compact tube in $z = (u, \theta, \pi_u, P)$, its Hamiltonian is

$$H_0 = a\zeta_0, \qquad \zeta_0 = \frac{1 + |P|^2_{\gamma^{-1}}}{2p}, \qquad p = -\pi_u - b \cdot P \geq p_0 > 0.$$

The small lapse is retained. The next quantitative implicit-root and Gronwall statement is local.

Proposition 6.1 (Root, gradient, and state budgets)**.** *Let the perturbed normalized unit shell on $|\zeta - \zeta_0| \leq b_0$ be*

$$F(v, z, \zeta) = 2p(v, z)\zeta - (1 + |P|^2_{\gamma^{-1}}) + e(v, z, \zeta) = 0.$$

Assume $p \geq p_0 > 0$, $|e| \leq \epsilon_0 < 2p_0 b_0$, $|e_\zeta| \leq \epsilon_\zeta \leq p_0$, $|D_z e| \leq \epsilon_1$, $|D_z p| \leq C_p$, and $|D_z \zeta_0| \leq C_0$. There is exactly one root in the strip, and

$$|\zeta - \zeta_0| \leq \epsilon_0/(2p_0), \tag{6.1}$$

$$|D_z\zeta - D_z\zeta_0| \leq C_p\epsilon_0/p_0^2 + (\epsilon_1 + \epsilon_\zeta C_0)/p_0. \tag{6.2}$$

For $H = a\zeta$, a Hamiltonian vector-field discrepancy bound is

$$\eta(v) = |D_z a| \, \frac{\epsilon_0}{2p_0} + a\left(\frac{C_p\epsilon_0}{p_0^2} + \frac{\epsilon_1 + \epsilon_\zeta C_0}{p_0}\right),$$

with bounds taken on the whole tube. If the background vector field has an integrable Lipschitz bound $L(v)$ there, on a common finite time interval, in either orientation, the state error is at most

$$e^{\int L}\left(\delta_{\text{initial}} + \int \eta(v)\,\mathrm{d}v\right). \tag{6.3}$$

A strict bound below the reference path's distance to the tube boundary certifies persistence in the tube.

Proof. The strip endpoint values of F have opposite strict signs, and $F_\zeta = 2p + e_\zeta \geq p_0$; this proves existence and uniqueness. At the root, $2p(\zeta - \zeta_0) = -e$, proving (6.1). Differentiation gives

$$(2p + e_\zeta)(D_z\zeta - D_z\zeta_0) = -2D_z p(\zeta - \zeta_0) - D_z e - e_\zeta D_z\zeta_0.$$

The bounds imply (6.2). Differentiate $a(\zeta - \zeta_0)$; the canonical symplectic matrix has Euclidean norm one. Comparing the two ODEs with the background Lipschitz bound and then applying Gronwall proves (6.3). A first-exit argument yields persistence. Smooth local vector fields and a tube on which the Lipschitz comparison is valid are assumed throughout. □

In the source, $L(v) \leq a(v)e^{o(v)}$ along the bounded-u branch is integrable, and the coefficient errors are exponentially small. These are inputs, not consequences of finite jet values. An entry hit additionally needs a transverse crossing-speed bound and a collar. The full-data seed and Cauchy attachment are separate. An elliptic Green inverse is not a hyperbolic propagator.

7 Comparison, Limitations, and Further Work

7.1 What is and is not improved

The source already provides the finite wave/gauge hierarchy, complex beams through real caustics, a common-rate triangular argument, exact constraint correction, and the order of the asymptotic choices. None of these is counted as a new theorem here. The new local assembly quantifies a geometric alternative to absolute center mass, retains its effect in the differentiated remainder and packet budgets, and supplies explicit finite error certificates.

The concentration gain is strongest on regular projections, fractional at the model fold, and absent for arbitrary collapsed center measures. It uses absolute integration and therefore cannot replace the sharper oscillatory L^2 theory valid through caustics. Its cost is a uniform *continuum* concentration bound on every computing slice, along with full differentiated amplitudes, physical coefficient norms, and imaginary-phase remainders. This report does not establish such a bound for every return-ray family in the source. Even at a fixed slice, a nondegenerate complex Hessian does not by itself imply a nondegenerate real center projection.

The sharp triangular factor reduces overestimation at fixed order. It does not improve the leading exponential rate of the source's qualitative construction. Likewise the phase and Hamiltonian tests certify only supplied finite arrays and tube bounds, not an entire infinite-time flow. There is no measured speedup and no historical priority claim for Gaussian integration, positive Volterra systems, the implicit-function theorem, or Gronwall.

The original private proofs [2, 3, 4] contribute the full coefficient-derivative, remainder, and perturbation discipline. Their finite inverse or weak elliptic statements do not imply the hyperbolic existence estimates used here. No older report's result or test count is recredited as a new M533 contribution.

7.2 Primary public context checked on October 9, 2026

Liu–Runborg–Tanushev [5] establish Gaussian beam superposition error estimates in well-posedness norms, with caustics included. Their result is not the same as our conditional absolute pointwise concentration bound. The primary abstract and version record were checked; the full proof was not independently audited here.

Häfner–Hintz–Vasy [6] address linear stability in the full subextremal Kerr range. Hintz's 2026 nonlinear preprint [7] uses structured asymptotics: a finite expansion of powers and logarithms with exponents greater than one, plus a faster-decaying remainder. Its exterior stability assertion is not the assigned source's arbitrary-symbol-tail statement about full-development C^2 future inextendibility. Dafermos–Luk [8] treat continuous extendibility across part of a Cauchy horizon from appropriate interior data. The distinction between C^0 and C^2, and between interior initial hypotheses and original bridge data, is essential. For these three records too, access was to primary abstracts and version/publication metadata, not full proofs.

7.3 Feasible extensions

A useful next step is a geometric concentration theorem for restricted return-ray families with uniform finite-type constants. It would need to track multiplicities, center density, moving charts, and possible higher caustics over the increasing time interval. An independent alternative is to use an oscillatory operator estimate in the precise physical mixed norms; cancellation would require more than the absolute measure bounds used here.

For certified finite implementations, one can propagate error intervals for H, B, imaginary Taylor remainders, and state-tube defects together with (5.2). These are proposals, not implemented Kerr simulations. The full symbol-tail evolution, derivative-dependent thresholds, nonlinear correction, and global Cauchy attachment remain the major imported obligations.

8 Verification and Evidence Boundaries

The public inventory contains 19 text files. Fresh literal reading covered the entire introduction, category section, 1,506-line packet section, and triangular appendix, as well as the root, macros, README, interior lines 779–1220, comparison lines 190–294 and 856–959, and constraint appendix lines 1–125. This totals 2,947 public lines. The remaining files and intervals are inventoried with explicit unread complements. Heading searches only locate architecture. The history fragment was not used as mathematical proof evidence.

Fresh private reading covered all 446 lines of the original local-moment chapter, the original jet section at lines 509–660 and 734–850, and the original stability section at lines 1–130: 845 lines in total. Included chapter files were not silently counted as read. Pre-read hashes were persisted before these displays; post-read hashes and captured-tree comparison test byte identity, not mathematical validity. No confidential material was sent to a public search.

The owned executable checks test exact rational Gaussian-moment coefficient relations, full derivative/Taylor identities, triangular powers and propagator composition, phase-block defects, normalized shell derivatives, the displayed packet margins, and the fold constant specialization. Cases and individual assertions are counted separately in the accompanying receipt: 203 parameter cases and 1,015 exact assertions passed, with no floating-point cases. They are manufactured algebraic fixtures, not interval-certified Kerr rays or Einstein evolutions. The proofs above establish the local statements; finite checks are supplemental. Parent checks are recorded separately and are not included in the owned counts.

9 Conclusion

The finite packet mechanism has several logically separate requirements: a physical wave energy, a center measure, full differentiated phase and amplitude remainders, a continued Hamiltonian branch, and a genuine constraint-corrected development. The conditional concentration gain and exact finite budgets strengthen the local part without pretending that any one of these certificates supplies the others. Applicability remains L3; the demonstrated C3 contribution is local, conditional, and independent of the source's global headline.

References

[1] OpenAI. *Quantitative Near-Kerr Evolution and Generic C^2 Future Inextendibility*, September 23, 2026. Pinned public source. Captured-tree identity is separate from proof verification.

[2] M. Subbarao. Original private *RamCalBooks*, volume 1-rev2, `ch03-local-moments-rev2.tex`, lines 1–446. Taylor integral remainders, coefficient derivatives, and polynomial closure distinctions; included chapter files not audited.

[3] M. Subbarao. Original private *RamNonlinearIE-PDE*, TR1-lin-nonlin-rev2, `sec-03-jets.tex`, lines 509–660 and 734–850. Full Leibniz rows and differentiated-remainder proofs.

[4] M. Subbarao. Original private *RamNonlin-IE-PDE-astra*, TR1-lin-nonlin, `stability.tex`, lines 1–130. Finite perturbation and boundary-aware weak response; not a hyperbolic propagation theorem.

[5] H. Liu, O. Runborg, N. M. Tanushev. *Error Estimates for Gaussian Beam Superpositions.* arXiv:1008.1320v2, June 2, 2011. Mathematics of Computation 82 (2013), 919–952.

[6] D. Häfner, P. Hintz, A. Vasy. *Linear stability of Kerr black holes in the full subextremal range.* arXiv:2506.21183v1, June 26, 2025.

[7] P. Hintz. *Nonlinear stability of subextremal Kerr black holes.* arXiv:2606.28253v2, August 3, 2026.

[8] M. Dafermos, J. Luk. *The interior of dynamical vacuum black holes I: The C^0-stability of the Kerr Cauchy horizon.* arXiv:1710.01722v1; Annals of Mathematics 202 (2025), 309–630.